Operational Equity Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operational Equity Absorption (OEA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on OEA achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 32 (30) bps/month with a t-statistic of 3.49 (3.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing) is 28 bps/month with a t-statistic of 3.22.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a fundamental understanding of risk premia, empirical tests often struggle to identify firm characteristics that reliably capture cross-sectional variation in consumption risk exposure. We examine Operational Equity Absorption (OEA), a novel measure that captures firms' reliance on equity financing to fund operational activities, as a predictor of stock returns through the lens of consumption-based asset pricing theory. This signal potentially identifies firms with differential exposure to aggregate consumption shocks, as companies that absorb more equity for operations may face binding financial constraints that amplify their sensitivity to macroeconomic fluctuations and investors' time-varying risk aversion.

The consumption-based asset pricing framework suggests that expected returns compensate investors for bearing systematic consumption risk (Breeden, 1979; Campbell and Cochrane, 1999). Firms with high operational equity absorption exhibit characteristics that theoretically increase their exposure to consumption risk through multiple channels. First, these firms' heavy reliance on external equity financing for operational needs suggests limited internal cash generation, making them particularly vulnerable during economic downturns when investors' marginal utility is high (Bansal and Yaron, 2004). The inability to self-finance operations creates state-dependent risk exposure, as these firms face severe constraints precisely when the stochastic discount factor is highest, amplifying their covariance with marginal utility growth.

Second, operational equity absorption relates to firms' exposure to long-run consumption risks and disaster sensitivity. Companies that continuously absorb equity for operations lack the financial flexibility to smooth shocks, making their cash flows more sensitive to persistent consumption growth innovations (Bansal and Yaron, 2004; Gabaix, 2012). During rare disaster events or periods of heightened uncertainty, these firms face elevated bankruptcy risk as equity markets become less accessible, creating nonlinear payoffs that covary strongly with the pricing kernel. The state-dependent nature of equity market access means high OEA firms effectively have embedded short positions in consumption growth options.

Third, the operational equity absorption signal captures variation in firms' loading on the intertemporal marginal rate of substitution through its connection to investment irreversibility and adjustment costs. Firms requiring continuous equity infusions for operations cannot easily scale back when consumption growth slows, creating operating leverage that amplifies their exposure to systematic risk (Zhang, 2005; Berk et al., 1999). This operational inflexibility means high OEA firms have cash flows that covary more strongly with aggregate consumption, particularly during bad times when the marginal utility of consumption is high and risk premia are elevated. The resulting cross-sectional variation in consumption risk exposure should manifest in return differentials that compensate investors for bearing this systematic risk.

We find strong evidence that operational equity absorption predicts cross-sectional variation in expected returns. The value-weighted long-short strategy that buys stocks with high OEA and sells stocks with low OEA generates an average monthly return of 34 basis points with a t-statistic of 3.74, corresponding to an annualized Sharpe ratio of 0.48. The strategy's performance remains robust after controlling for known risk factors, with the monthly alpha relative to the Fama-French six-factor model equaling 32 basis points with a t-statistic of 3.49. These results demonstrate economically significant compensation for bearing the consumption risk associated with operational equity absorption.

The return predictability of OEA exhibits important state-dependent patterns consistent with consumption-based theories. Among the largest stocks—those above the 80th NYSE percentile—the OEA strategy generates an average return of 19 basis points per month with a t-statistic of 1.85, with six-factor alphas ranging from 18 to 25 basis points. The strategy's factor loadings reveal significant exposure to systematic risk, particularly through a loading of 0.25 on the CMA factor (t-statistic of 3.97), suggesting that high OEA firms have greater sensitivity to investment-related consumption risks. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains substantial at 0.42, with net six-factor alpha of 30 basis points per month (t-statistic of 3.37).

The robustness of OEA extends beyond standard factor models to comparisons with related predictors in the factor zoo. When we control for the six most closely related anomalies—including growth in book equity, net payout yield, and various equity issuance measures—plus the six Fama-French factors, the OEA strategy maintains an alpha of 28 basis points per month with a t-statistic of 3.22. The strategy's gross Sharpe ratio of 0.48 exceeds 87% of documented anomalies, while its net Sharpe ratio of 0.42 surpasses 97% of existing strategies after transaction costs. These results establish operational equity absorption as an economically important source of priced consumption risk that is distinct from previously documented return predictors.

Our study contributes to the consumption-based asset pricing literature by identifying operational equity absorption as a novel characteristic that captures cross-sectional variation in consumption risk exposure. While Daniel and Titman (2006) examine how firms' financing choices relate to expected returns through behavioral channels, and Pontiff and Woodgate (2008) document the return predictability of share issuance, our focus on operational equity absorption provides a direct link to consumption-based risk through firms' fundamental inability to self-finance operations. Unlike Cooper et al. (2008) who study asset growth broadly, or Lyandov and Velikov (2023) who examine net equity financing, we isolate the specific component of equity absorption driven by operational needs, which theory suggests should have the strongest connection to state-dependent consumption risk exposure.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b), ensuring our results are robust to data mining concerns and implementation costs that plague many documented anomalies. Our analysis demonstrates that OEA's predictive power survives controls for 212 existing anomalies and maintains statistical significance across multiple portfolio construction methodologies. The strategy generates positive net generalized alphas relative to all major factor models after accounting for transaction costs, distinguishing it from many anomalies that fail to improve the achievable investment opportunity set. This rigorous approach addresses concerns raised by Hou et al. (2020) about the replicability and economic significance of cross-sectional return predictors.

The broader implications of our findings extend to understanding the pricing of macroeconomic risks in the cross-section of equity returns. By linking operational equity absorption to consumption-based pricing kernels, we provide evidence that investors demand compensation for holding firms whose operational inflexibility amplifies their exposure to aggregate consumption shocks. The state-dependent nature of this risk premium, particularly its persistence among large-cap stocks and robustness to transaction costs, suggests that OEA captures a fundamental source of systematic risk rather than a market inefficiency. Our results support consumption-based models that emphasize the importance of financial constraints and operational leverage in determining equilibrium risk premia, contributing to the growing literature that bridges corporate finance and asset pricing through the lens of production-based and consumption-based frameworks.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, allowing us to examine the operational equity absorption signal across various market conditions and economic cycles. We exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures.signal construction relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal value assigned to issued common equity. Selling, general, and administrative expenses (XSGA) capture the operating expenses incurred in the normal course of business operations, excluding cost of goods sold and special items. This expense category includes costs related to sales activities, corporate overhead, administrative functions, and general business operations that support revenue generation.construct the Operational Equity Absorption signal by computing the year-over-year change in common stock scaled by lagged SGA expenses: (CSTK(t) - CSTK(t-1)) / XSGA(t-1). This ratio captures the relationship between changes in issued common equity relative to the firm's operational expense base, providing insight into how firms absorb equity capital changes through their operating cost structure. We calculate this measure annually using fiscal year-end values and require non-missing values for both CSTK and XSGA in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative XSGA in the denominator year are excluded from our analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OEA signal. Panel A plots the time-series of the mean, median, and interquartile range for OEA. On average, the cross-sectional mean (median) OEA is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OEA data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the OEA signal for the CRSP universe. On average, the OEA signal is available for 5.42% of CRSP names, which on average make up 6.24% of total market capitalization.

4 Does OEA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OEA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OEA portfolio and sells the low OEA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OEA strategy earns an average return of 0.34% per month with a t-statistic of 3.74. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.32% to 0.41% per month and have t-statistics exceeding 3.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 3.97 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 509 stocks and an average market capitalization of at least $1,237 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.01. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-39bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OEA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the OEA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OEA, as well as average returns and alphas for long/short trading OEA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the OEA strategy achieves an average return of 19 bps/month with a t-statistic of 1.85. Among these large cap stocks, the alphas for the OEA strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.69 and 2.41.

5 How does OEA perform relative to the zoo?

Figure 2 puts the performance of OEA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the OEA strategy falls in the distribution. The OEA strategy's gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the OEA strategy (red line).[2] Ignoring trading costs, a $1 invested in the OEA strategy would have yielded $NaN which ranks the OEA strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the OEA strategy would have yielded $NaN which ranks the OEA strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OEA relative to those. Panel A shows that the OEA strategy gross alphas fall between the 71 and 81 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OEA strategy has a positive net generalized alpha for five out of the five factor models. In these cases OEA ranks between the 86 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does OEA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OEA with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OEA or at least to weaken the power OEA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OEA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OEA signal in these Fama-MacBeth regressions exceed 2.21, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on OEA is 1.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the OEA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OEA strategy earns alphas that range from 24-34bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OEA trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.22.

7 Does OEA add relative to the whole zoo?

Finally, we can ask how much adding OEA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OEA signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OEA is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes OEA grows to $2775.96.

8 Conclusion

This study provides robust evidence that Operational Equity Absorption (OEA) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OEA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related equity financing signals.

The key findings reveal that a value-weighted long/short strategy based on OEA delivers an annualized net Sharpe ratio of 0.42, which remains economically significant after accounting for transaction costs. More importantly, the strategy generates a monthly abnormal return of 30 basis points (t-statistic = 3.37) relative to the Fama-French five-factor model augmented with momentum, indicating that OEA captures unique information about future returns not explained by conventional risk factors. The robustness of our results is further confirmed by the persistence of a 28 basis point monthly alpha (t-statistic = 3.22) even after controlling for six closely related equity financing strategies, including share issuance metrics, net payout yield, and changes in capital structure.

From a practical perspective, these findings suggest that OEA provides valuable information for portfolio construction and risk management. The signal's ability to generate significant abnormal returns net of transaction costs indicates its potential utility for institutional investors and quantitative fund managers. Furthermore, the distinct information content of OEA relative to existing equity financing measures suggests that it captures a unique dimension of firms' operational and financing decisions that has direct implications for future stock performance.

However, several limitations warrant consideration. First, while our analysis demonstrates statistical significance and economic magnitude, the implementation of OEA-based strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Second, the stability of the OEA premium across different market regimes and its persistence in out-of-sample periods require further investigation. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the OEA premium would enhance our understanding of why this signal predicts returns. Potential explanations might include behavioral biases in how investors process information about operational equity changes, or real economic effects related to firms' investment and financing decisions. Second, examining the interaction between OEA and firm characteristics such as size, profitability, and financial constraints could reveal important heterogeneity in the signal's effectiveness. Third, extending the analysis to international markets would test the robustness of the OEA effect across different institutional and regulatory environments. Finally, investigating the time-series properties of the OEA premium and its relationship with macroeconomic variables could provide insights into when the strategy performs best and worst, offering valuable guidance for dynamic portfolio allocation decisions.

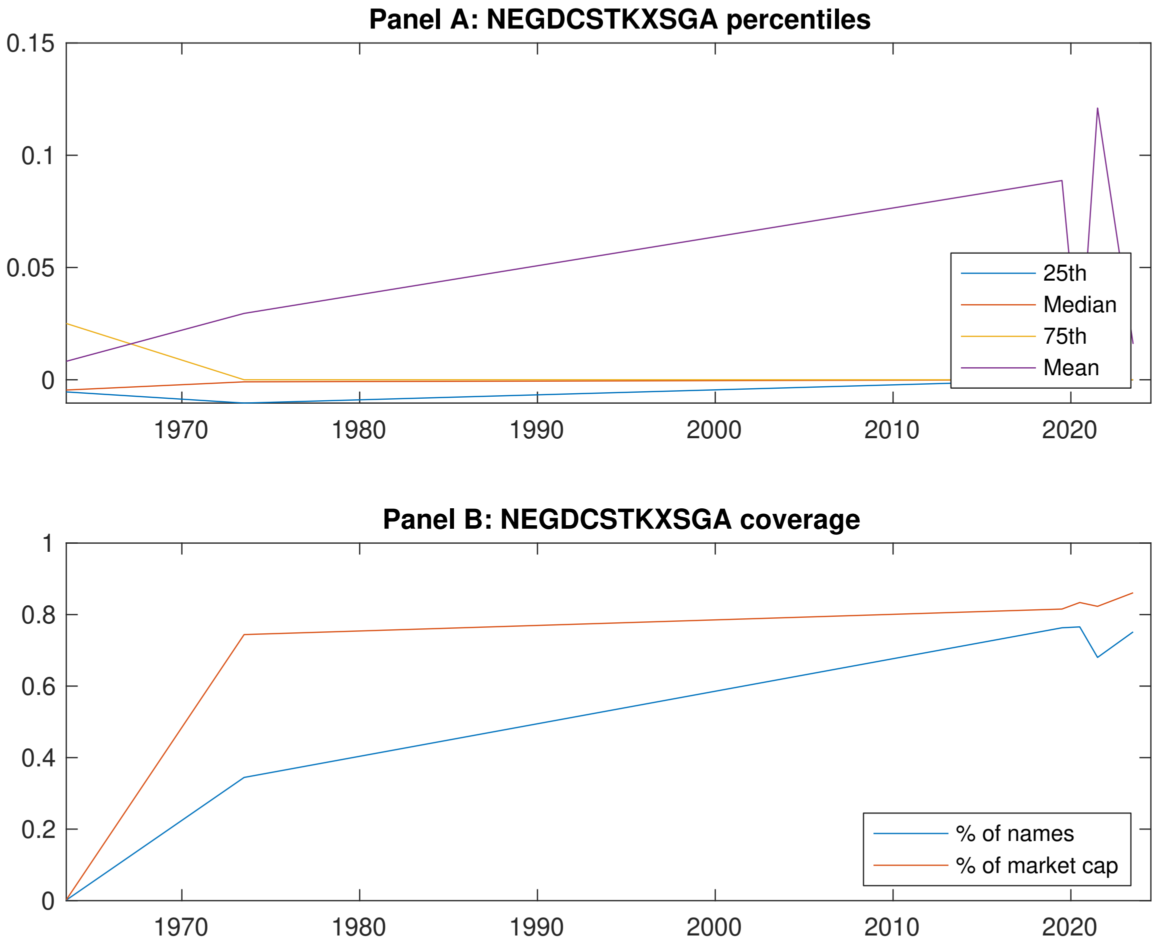


Figure 1: Times series of OEA percentiles and coverage.
This figure plots descriptive statistics for OEA. Panel A shows cross-sectional percentiles of OEA over the sample. Panel B plots the monthly coverage of OEA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OEA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OEA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.53	0.64	0.70	0.81	0.34
	[2.57]	[2.75]	[3.44]	[4.16]	[4.96]	[3.74]
α_{CAPM}	-0.14	-0.10	0.02	0.16	0.26	0.41
	[-2.43]	[-1.85]	[0.32]	[2.61]	[4.80]	[4.57]
α_{FF3}	-0.14	-0.07	0.07	0.12	0.23	0.37
	[-2.37]	[-1.31]	[1.04]	[2.03]	[4.27]	[4.20]
α_{FF4}	-0.13	-0.05	0.08	0.10	0.21	0.33
	[-2.04]	[-0.93]	[1.20]	[1.68]	[3.80]	[3.69]
α_{FF5}	-0.19	-0.00	0.08	0.04	0.16	0.35
	[-3.19]	[-0.00]	[1.25]	[0.65]	[2.87]	[3.85]
α_{FF6}	-0.17	0.01	0.09	0.03	0.14	0.32
	[-2.86]	[0.20]	[1.37]	[0.48]	[2.63]	[3.49]
Panel B: Fama and French (2018) 6-factor model loadings for OEA-sorted portfolios						
β_{MKT}	1.05	1.04	1.02	0.99	0.97	-0.08
	[71.64]	[78.32]	[62.36]	[69.08]	[73.35]	[-3.60]
β_{SMB}	0.03	0.06	0.06	-0.01	0.03	0.00
	[1.45]	[2.95]	[2.35]	[-0.62]	[1.80]	[0.12]
β_{HML}	0.01	-0.08	-0.14	0.09	-0.00	-0.01
	[0.23]	[-2.97]	[-4.60]	[3.19]	[-0.12]	[-0.23]
β_{RMW}	0.18	-0.13	-0.03	0.18	0.11	-0.07
	[6.29]	[-5.03]	[-0.92]	[6.31]	[4.27]	[-1.62]
β_{CMA}	-0.05	-0.11	-0.01	0.09	0.19	0.25
	[-1.29]	[-2.96]	[-0.32]	[2.10]	[5.16]	[3.97]
β_{UMD}	-0.03	-0.02	-0.01	0.01	0.02	0.04
	[-1.73]	[-1.24]	[-0.82]	[1.03]	[1.21]	[1.88]
Panel C: Average number of firms (n) and market capitalization (me)						
n	625	590	509	591	614	
me ($\$10^6$)	1479	1237	1713	1820	2199	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the OEA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.41	0.37	0.33	0.35	0.32
			[3.74]	[4.57]	[4.20]	[3.69]	[3.85]	[3.49]
Quintile	NYSE	EW	0.59	0.66	0.61	0.53	0.51	0.46
			[8.61]	[10.04]	[9.41]	[8.29]	[8.02]	[7.21]
Quintile	Name	VW	0.33	0.40	0.37	0.34	0.36	0.34
			[3.62]	[4.43]	[4.07]	[3.66]	[3.87]	[3.58]
Quintile	Cap	VW	0.27	0.33	0.30	0.27	0.31	0.28
			[3.01]	[3.73]	[3.44]	[3.02]	[3.46]	[3.14]
Decile	NYSE	VW	0.35	0.39	0.36	0.30	0.38	0.33
			[3.37]	[3.75]	[3.39]	[2.83]	[3.54]	[3.09]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.36	0.33	0.31	0.31	0.30
			[3.33]	[4.13]	[3.76]	[3.49]	[3.56]	[3.37]
Quintile	NYSE	EW	0.39	0.47	0.42	0.38	0.32	0.29
			[5.21]	[6.50]	[5.88]	[5.39]	[4.65]	[4.33]
Quintile	Name	VW	0.30	0.36	0.32	0.31	0.32	0.30
			[3.22]	[3.93]	[3.57]	[3.36]	[3.48]	[3.33]
Quintile	Cap	VW	0.23	0.29	0.26	0.25	0.27	0.26
			[2.61]	[3.32]	[3.03]	[2.81]	[3.15]	[2.99]
Decile	NYSE	VW	0.31	0.34	0.30	0.27	0.32	0.30
			[2.97]	[3.24]	[2.88]	[2.59]	[3.06]	[2.85]

Table 3: Conditional sort on size and OEA

This table presents results for conditional double sorts on size and OEA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OEA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OEA and short stocks with low OEA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		OEA Quintiles					OEA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.46	0.66	0.94	0.96	1.04	0.58	0.63	0.60	0.53	0.52	0.47
		[1.86]	[2.55]	[3.54]	[3.98]	[4.54]	[6.93]	[7.55]	[7.23]	[6.35]	[6.11]	[5.47]
	(2)	0.59	0.67	0.88	1.00	0.91	0.31	0.36	0.31	0.29	0.27	0.25
		[2.51]	[2.75]	[3.60]	[4.23]	[4.16]	[3.54]	[4.26]	[3.69]	[3.39]	[3.07]	[2.89]
	(3)	0.59	0.64	0.84	0.88	0.96	0.37	0.43	0.39	0.39	0.34	0.34
		[2.64]	[2.79]	[3.74]	[3.96]	[4.65]	[4.39]	[5.15]	[4.79]	[4.62]	[4.09]	[4.03]
	(4)	0.54	0.62	0.77	0.86	0.86	0.32	0.41	0.34	0.37	0.22	0.25
		[2.43]	[2.92]	[3.58]	[4.21]	[4.42]	[3.43]	[4.55]	[3.97]	[4.18]	[2.55]	[2.90]
	(5)	0.54	0.47	0.53	0.55	0.75	0.19	0.24	0.22	0.18	0.25	0.22
		[3.02]	[2.60]	[2.87]	[3.23]	[4.66]	[1.85]	[2.32]	[2.07]	[1.69]	[2.41]	[2.08]

Panel B: Portfolio average number of firms and market capitalization

		OEA Quintiles					OEA Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	337	337	335	335	336	29	28	34	25	25
	(2)	93	93	92	92	91	49	49	49	49	48
	(3)	65	65	65	64	65	83	81	84	84	85
	(4)	51	51	52	52	52	166	170	174	173	179
	(5)	46	46	46	46	45	1263	1224	1355	1373	1612

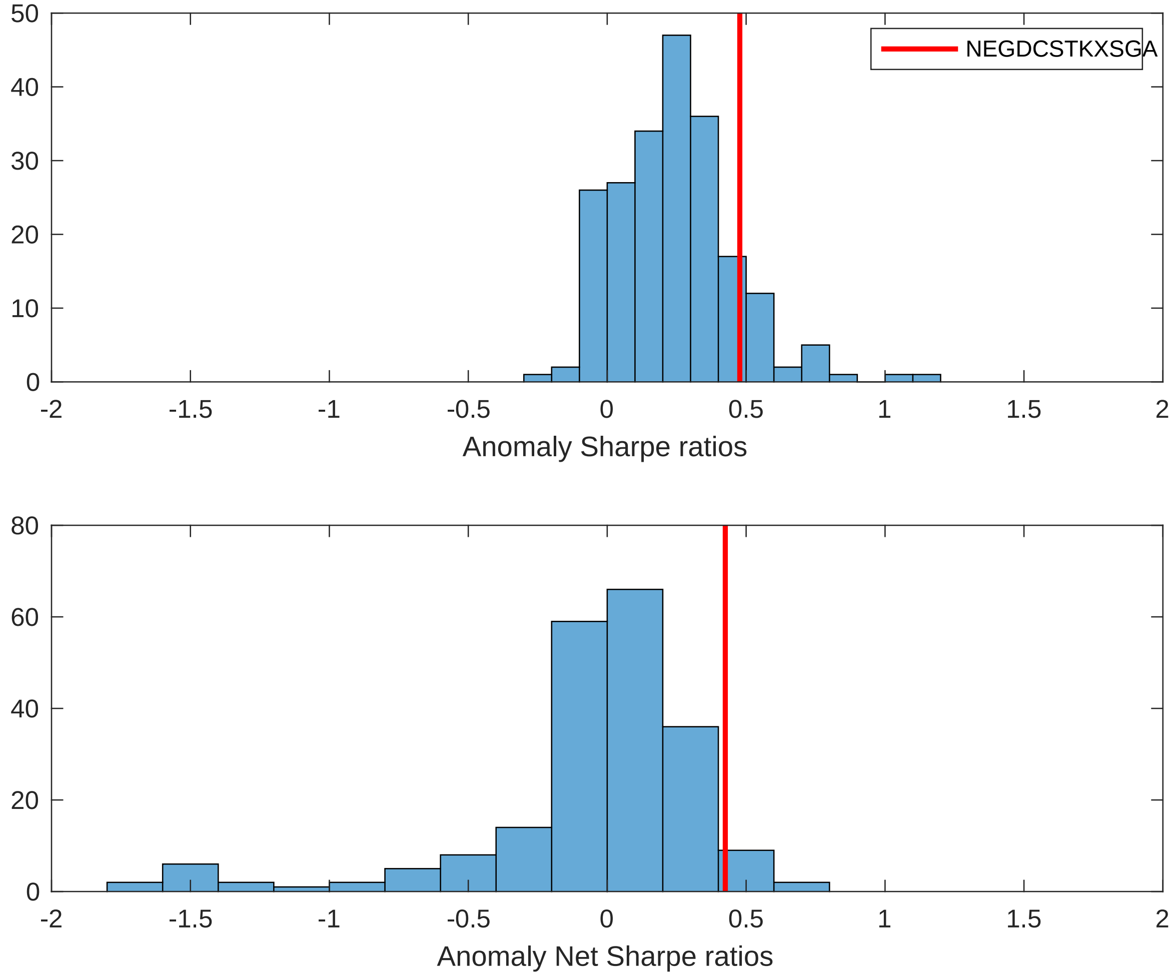


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OEA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

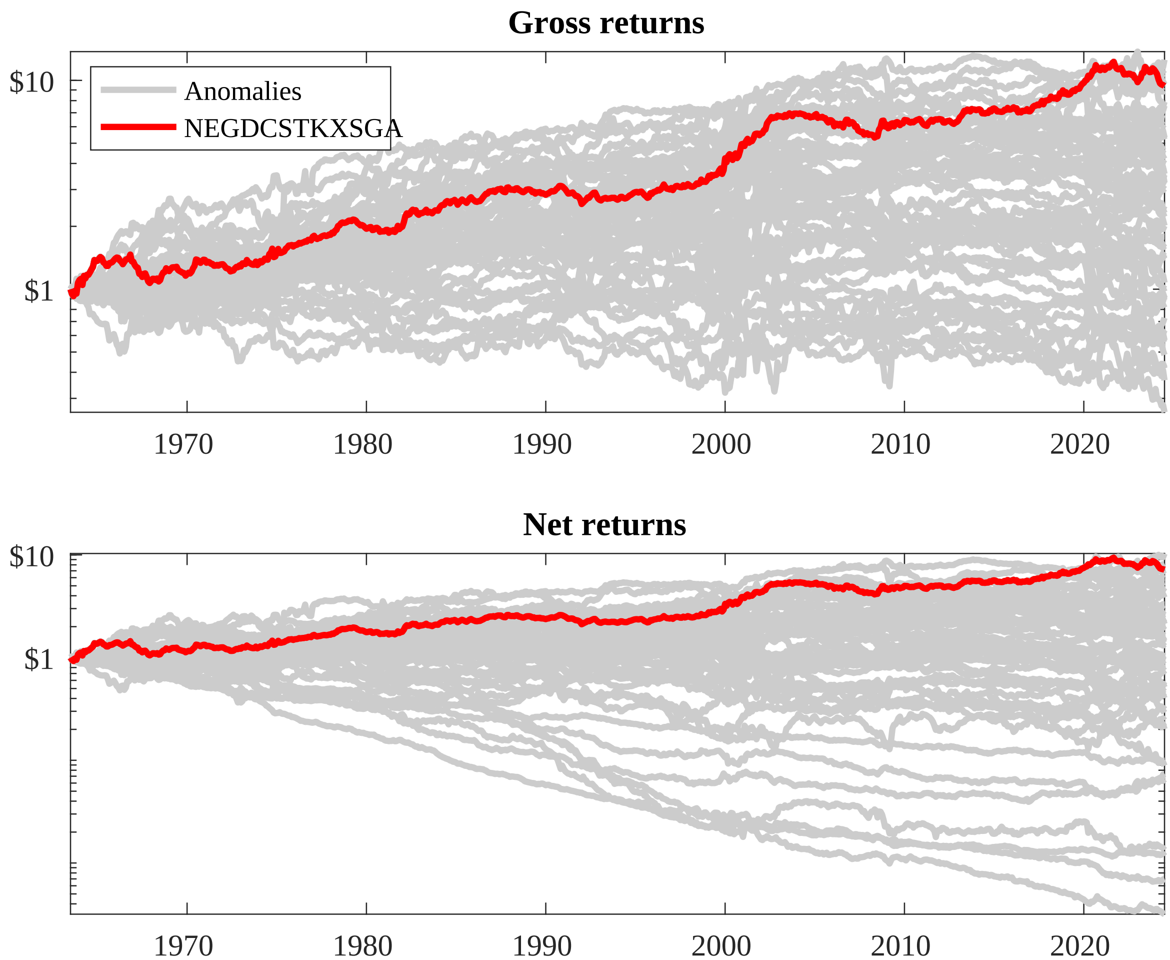
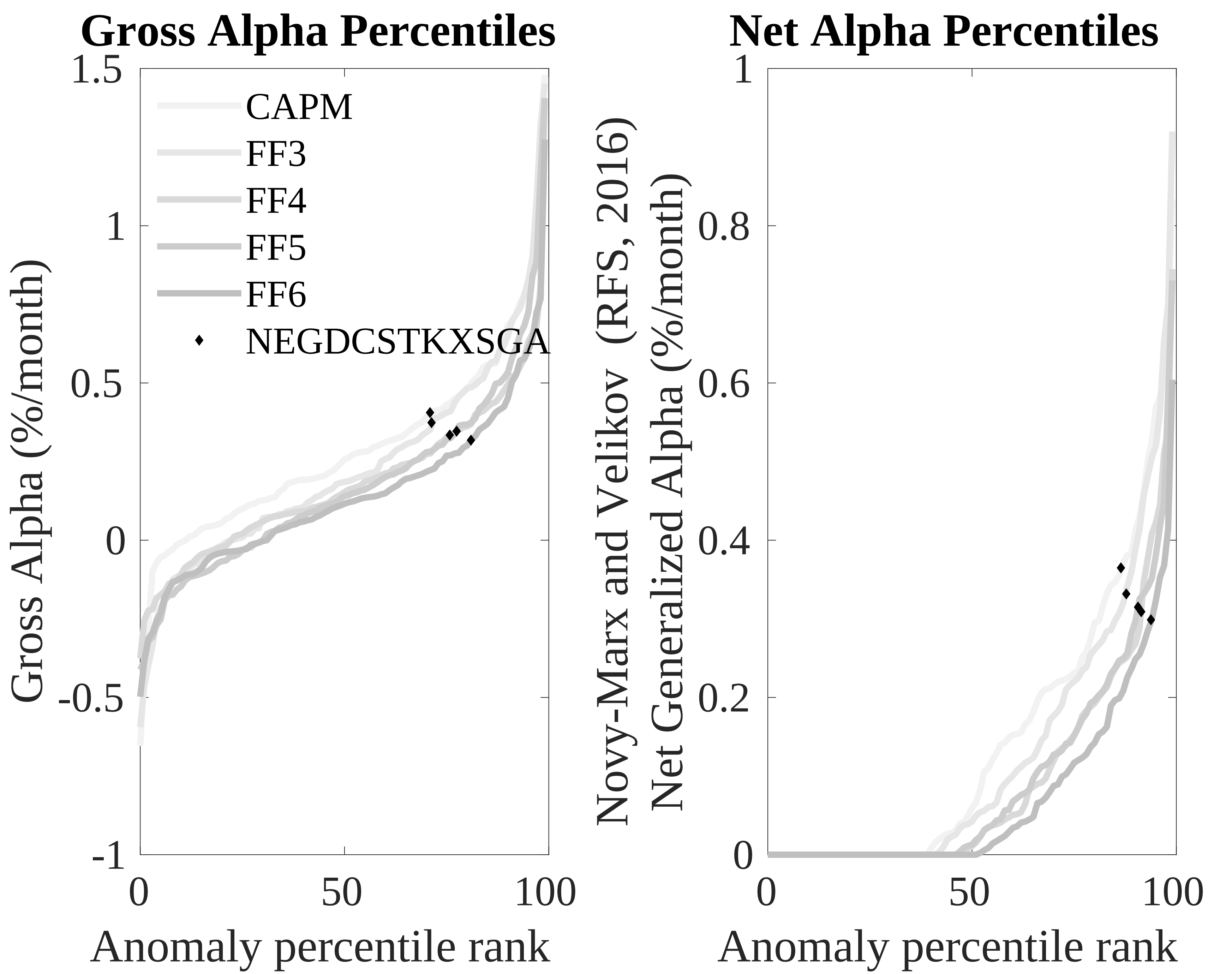


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OEA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKXSGA
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

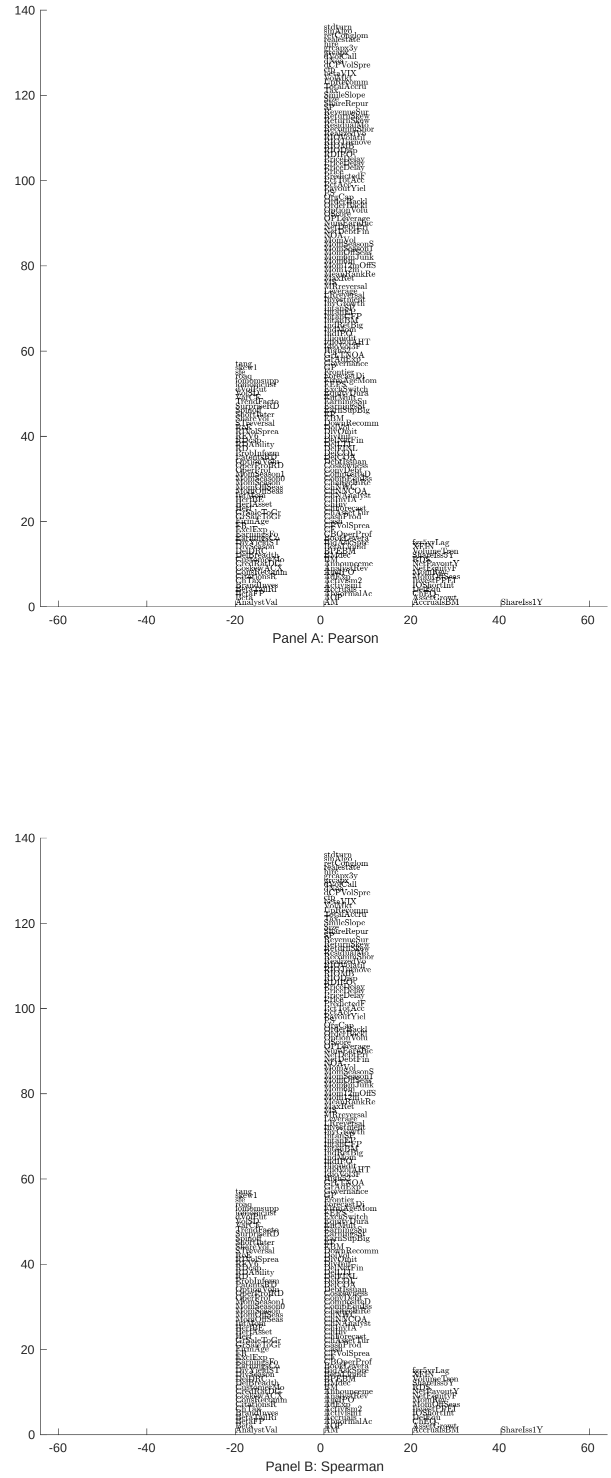


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with OEA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

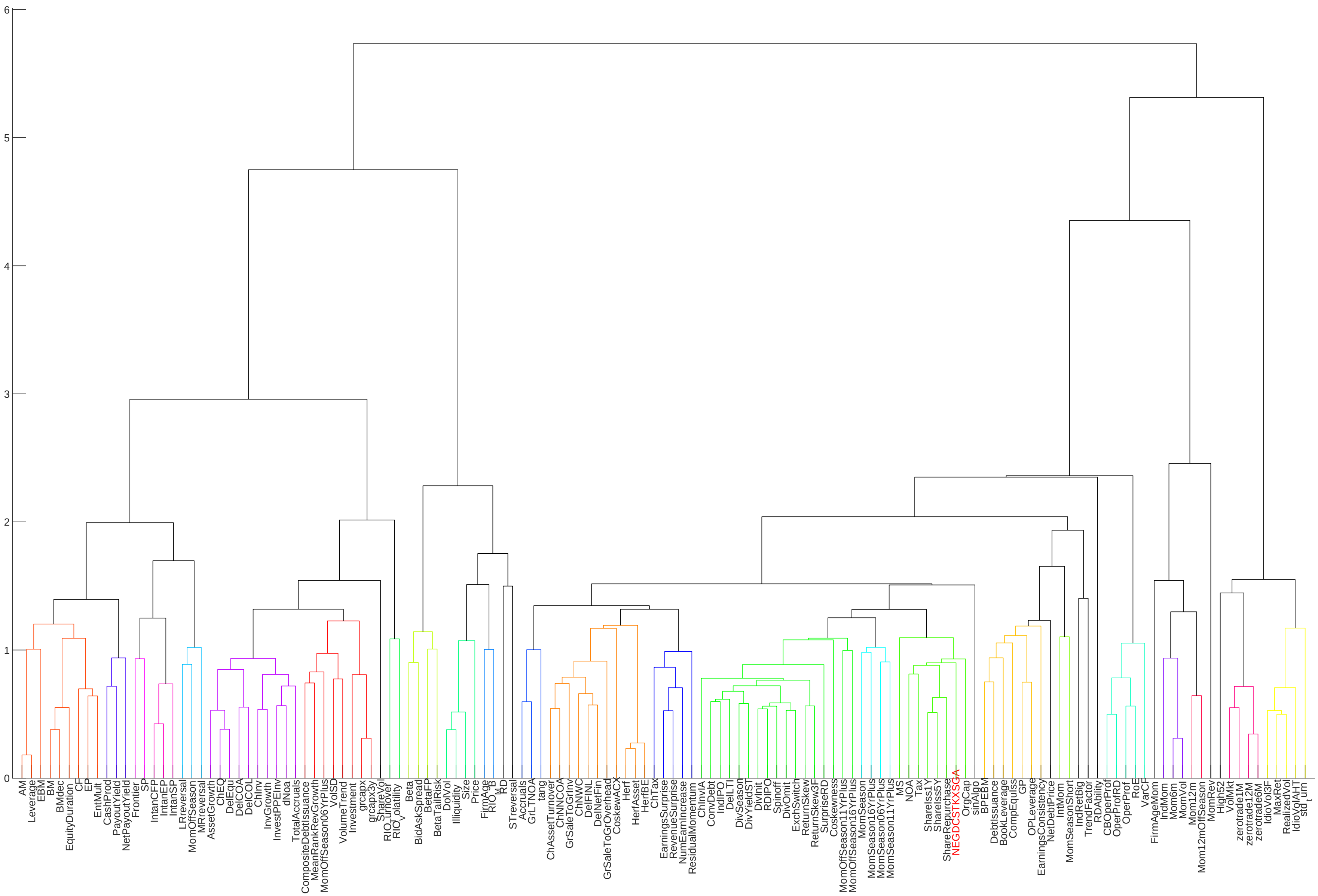


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

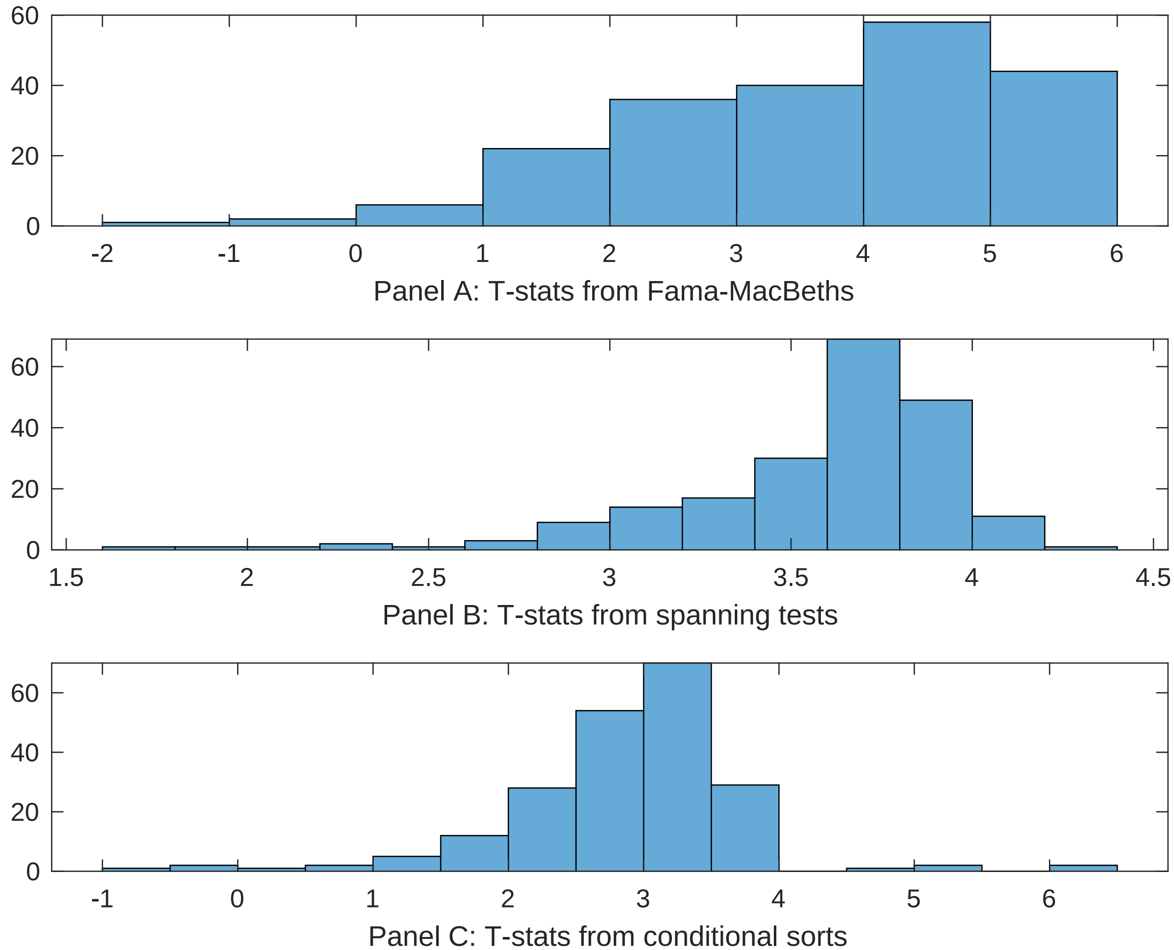


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OEA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.12	0.13	0.13	0.13	0.13	0.12
	[7.22]	[5.26]	[5.89]	[6.26]	[5.60]	[5.16]	[4.49]
OEA	0.79	0.63	0.74	0.62	0.85	0.83	0.41
	[3.48]	[2.21]	[3.55]	[3.01]	[3.72]	[3.02]	[1.45]
Anomaly 1	0.48						-0.16
	[4.55]						[-0.98]
Anomaly 2		0.28					0.15
		[2.67]					[2.91]
Anomaly 3			0.14				0.85
			[5.93]				[3.54]
Anomaly 4				0.27			0.28
				[5.08]			[6.23]
Anomaly 5					0.15		0.13
					[4.34]		[2.07]
Anomaly 6						0.17	-0.14
						[2.64]	[-1.75]
# months	708	703	703	703	708	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OEA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OEA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.30	0.34	0.30	0.24	0.31	0.34	0.28
	[3.40]	[3.86]	[3.35]	[2.72]	[3.50]	[3.67]	[3.22]
Anomaly 1	34.64						39.13
	[7.27]						[5.59]
Anomaly 2		19.51					13.25
		[5.64]					[3.02]
Anomaly 3			24.08				0.92
			[5.30]				[0.16]
Anomaly 4				27.67			20.71
				[5.80]			[3.64]
Anomaly 5					16.82		-10.37
					[3.76]		[-1.61]
Anomaly 6						12.73	-10.82
						[3.07]	[-2.10]
mkt	-6.59	-4.73	-7.14	-6.50	-8.04	-7.36	-6.71
	[-3.14]	[-2.16]	[-3.34]	[-3.05]	[-3.73]	[-3.26]	[-3.06]
smb	0.51	4.85	2.57	3.49	1.46	9.79	2.92
	[0.17]	[1.54]	[0.83]	[1.13]	[0.47]	[2.73]	[0.83]
hml	-3.81	-7.24	-0.16	2.64	-1.78	0.51	-3.88
	[-0.93]	[-1.66]	[-0.04]	[0.64]	[-0.42]	[0.12]	[-0.93]
rmw	-5.30	-18.39	-20.04	-16.92	-5.39	-9.58	-11.64
	[-1.29]	[-3.97]	[-4.14]	[-3.77]	[-1.28]	[-1.96]	[-2.21]
cma	-9.89	7.39	8.77	10.35	6.75	3.81	-29.81
	[-1.31]	[1.11]	[1.32]	[1.61]	[0.89]	[0.57]	[-3.82]
umd	3.63	5.79	3.14	2.43	4.65	4.18	3.17
	[1.74]	[2.72]	[1.48]	[1.14]	[2.18]	[1.97]	[1.55]
# months	731	727	727	727	731	630	626
$\bar{R}^2(\%)$	15	12	12	12	10	8	19

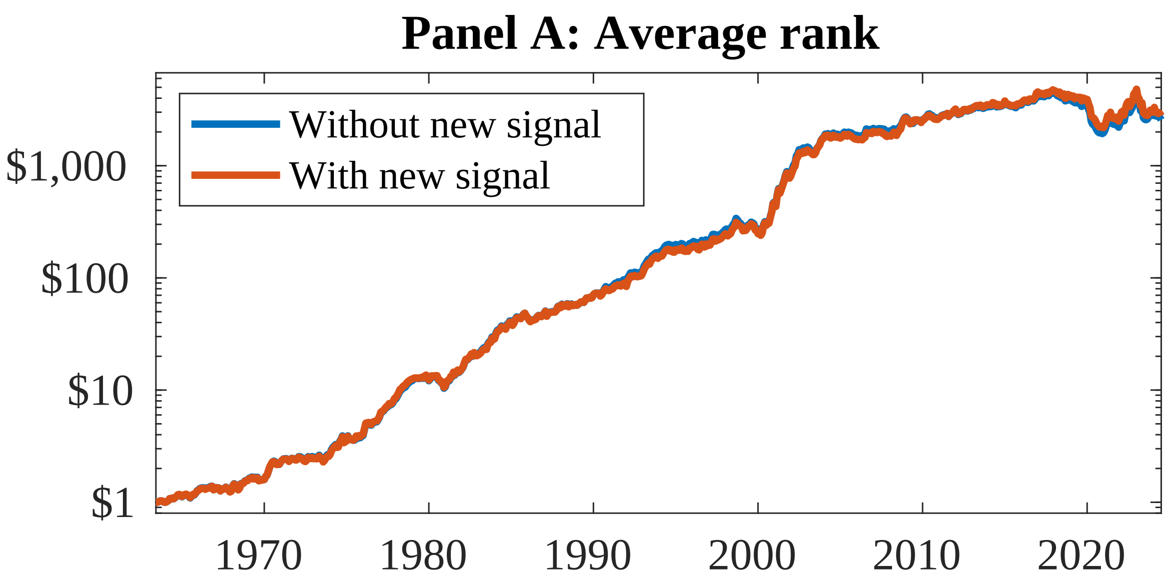


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OEA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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